What explains IPO survival?: A global perspective

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Abstract

This international study examines the determinants of the IPO survival rates using firm-level characteristics, market development and stakeholder protection. The study covers 7,362 IPOs listed between 2000 and 2008 across thirty-two countries. Our results show that the impact of firm's characteristics on the survival rates vary by region of the IPO firms, although offer-size and profitability are the common determinants of high survival rates' post-listing. Likewise, domestic market liquidity and level of shareholder protection accelerate post-listing survival rates globally. Nonetheless, market and regulatory attribute's impact on survival rates differently based on the region in which the IPO firm is listed. Our findings are robust to a battery of sensitivity analysis, number of alternative model specifications, clustering and sample decompositions.

JEL clasificación: G15, G3, F3

Keywords: IPO, survival rate, market development, governance, international markets.

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1. Introduction

Over the last two decades, the integration of financial markets has attracted a considerable attention of academics and investors alike. For instance, prior studies have examined the extent to which European debt and equity markets are integrated (Baele et al., 2004). Investors have been keen on understanding how financial markets are correlated, and how country-specific shocks are transmitted from one market to another. An insight into these issues provides a crucial risk-hedging strategy *via* international diversification. Some studies have examined the correlation and co-movement among different markets (Erb et al., 1994; Karolyi and Stulz, 1996), while others have focused on risk spill-over between the markets. In general, the developed markets have documented a high degree of stock market integration (Friedman and Shachmurove, 1997; Huyghebaert and Quan, 2006). However, emerging stock markets are arguably less integrated internationally (Huyghebaert and Wang, 2010). The gap in this strand of literature is the profile of firms seeking listing in the stock market. The number and volume of firms seeking Initial Public Offering (*hereafter* IPO) are vital for the market development. A recent report by OECD shows that a well-functioning IPO market could deliver thousands of new jobs to the economy (Weild et al., 2013). Fama and French (2004) document that IPO market is seen as a key to the development of the stock market. The IPO market is likely to grow if the listing firms have the prospect of higher survival post-listing.[1] IPO firms are not only subject to the short-term risk of failure due to organizational transformation from private to public enterprise but also face a complex dilemma in terms of valuation due to high information asymmetry (Lowry et al., 2010). These changes can be hazardous to firms, especially those struggling to adapt their strategies and internal operations to be compatible with their new found public status.

[1] Unlike the large Financial Institutes with a '*Too big to fail'* resolution, IPO firms face a relatively higher risk of bankruptcy and failure (see detailed discussions of *Too big to fail* resolutions in Kaufman, 2014).

There are practical implications of IPO failures for all market participants. For instance, during the 1997-98 glory years of *Nouveau Marche*, hundreds of young French firms conducted IPOs followed by the virtual collapse of the French IPO market post March 2000 leading to a significant loss of investors' wealth. Moreover, post *dotcom* bubble and bankruptcy of the tech- IPO firms, capital markets have experienced puzzling episodes of crisis.[2] Regulators, competitors, creditors, and potential investors (both minority- and block-shareholders) analyse the IPO firms with the view to assess their long term viability.[3] Firm owners and managers desire survival and strong performance (Hensler et al., 1997). The incentive of investing in firms with high survival is common across all the markets. The firm survival provides an in-depth insight of the long-run performance, which is interesting to both long-term and short term investors as well as the stock market regulators.

In this research paper, we examine the survival rates of the newly listed firms across various markets over the contemporary time interval. This provides a better assessment of the effects of firms' characteristics, market development and stakeholder protection on the survival rates. Most of the prior research has focused on a single market, while we aim to provide international evidence on firms' survivals and highlight factors that are unique to the region of the IPO firms *vs.* common factors. An insight into the survival rates globally provides investors useful information about the kind of firms that are suitable candidates for long-term investments, in particular, for those planning to commit capital outside their home market *viz.* ADRs and GDRs.

Our results show that the survival rates are comparable and high in the short-run, *i.e.* three-years post-listing for firms listed in the North-America, Europe, BRICS, and East-Asia & Australia. However, over a five-year period post-listing, firms listed in Europe have the

[2] See Craig and Putka (2002), (Wall Street Journal, June 13, 2002) – 'Robertson Stephens, the once red-hot investment bank boutique that took some of the industry's best known dot-coms public only to see them founder, is itself now struggling to survive…….'
[3] Stock market investors are also concerned with the Seasoned Equity Offerings (SEOs) and announcement of large capital infusions like government bailout and private equity placement (Elyasiani et al., 2014).

lowest survival rates, while firms listed in East-Asia & Australia have the highest survival time. We find that the impact of firms' characteristics on the survival rates differs by region of the IPO firms. For instance, initial returns accelerate the survival time only in Europe, while leverage decelerates the survival time in East-Asia & Australia. Neither the survival rates of IPOs listed in North America, nor those listed in BRICS are influenced by initial returns or leverage. So the impact of firm's characteristics is not uniform and limited to the region of the IPO firm. Size and level of profitability at the time of listing accelerate the survival rate in all regions. Furthermore, local stock exchange's liquidity and domestic anti-director minority shareholder rights enhance the survival time post-listing. The importance of market liquidity and shareholder's rights to the IPO survivals has been overlooked in the extant literature. Other factors such as GDP per capita, stock market size and bond holders' rights are unique to the region in which the firm is listed. Together, our results show that IPO survivals are not only determined by firm characteristics as documented in the previous studies, but also by region's market conditions and governance factors.

This study contributes to the rapidly growing literature on globalisation and the IPOs by investigating the survival rates of the newly listed firms from a global perspective. First, we demonstrate common and unique IPO characteristics that accelerate / decelerate the survival rates globally. Second, we shed light on the importance of the level of market development and local governance factors on the survival rates. Third, we provide valuable insights to IPO investors planning to expand their investments outside their domestic market by highlighting differences in the survival rates across various regions. Finally, our results are useful to policy-makers who are keen on maintaining consistency or improving the survival rates of firms listed in their markets against those listed elsewhere.

The article proceeds as follows. Section 2 provides an overview of the literature and also details the hypothesis being tested; sample selection and methodology are discussed in Section 3. Section 4 reports descriptive statistics and empirical results, robustness checks are discussed in section 5, and section 6 concludes the paper.

2. Literature review and hypothesis development

2.1 Past literature on IPOs survivals

There is a large body of literature on the survival of newly listed stocks in the U.S. followed by a few on the U.K. stock market. The failure rates reported in the previous U.S. studies ranges from 2 to 9 percent over the first-year of listing, 6 to 42 percent over two-years, 9 to 47 percent over five-years and up to 58.50 percent over ten-years in the post-IPO period. Hensler et al. (1997) investigate the relation between the survival rate of the IPOs and firm characteristics using a hazard model. Their findings show that survival rates are positively related to firm age and size, IPO initial returns and insider ownership. Jain and Kini (1999) examine the probability of surviving post-IPO using a multinomial logit model. They find that firm size at the time of the IPO, pre-IPO operating performance and investment bankers' prestige are positively related to IPO survival. While in their follow-up study, Jain and Kini (2000) examine whether venture capital involvement improves the survival profile of the IPO firms. Their findings indicate that the probability of a post-IPO survival is significantly positively affected by the venture capital backing and the prestige of the investment bank leading the underwriting syndicate for the IPO. Jain and Martin (2005) investigate the relationship between audit quality and post-IPO survival using a proportional hazard model. They report that the hazard rate is negatively related to the auditor quality in the post-IPO era of the firm. Kooli and Meknassi (2007) examine the survival profile of the U.S. IPO issuers from 1985 to 2005. They find that large IPOs have lower probability of failing relative to small IPOs. They also find evidence that IPO underpricing increases the likelihood of failure, while having a prestigious underwriter improves the survivability of the issuing firm.

Fama and French (2004) examine ten-year post-issue survival rates of the U.S. *new lists* coming to the market between 1973 and 1991. They find that the characteristics of the companies going public, such as profitability and growth opportunity, significantly changed over their three-decade sample period. Authors noted that the more recent listings happen to exhibit lower profitability and higher growth potential. These changes in firm characteristics are associated with a sharp decline in survival rates of new lists. Fama and French (2004) argue that their results depict the changes in the characteristics and survival rates of new lists primarily due to a decline in the cost of equity; thereby allowing younger and less profitable firms to easily go public. Espenlaub et al. (2012) examine the survival rates of the firms listed on the U.K. Alternate Investment Market (AIM) and the role of the Nominated Advisor. They report that the survival rate of the AIM IPOs is comparable to the well-established OTC markets such as NASDAQ and the U.K. main market despite AIM being a second tier market. While Ahmed and Jelic (2014) recently documented that the survival rates are higher when the lock-up periods are longer post-listing.

However, these studies are specific to either North-America (*i.e.* the U.S. and Canada) or Europe (primarily U.K.), while the current study examines the survival rates of the IPOs globally. Factors influencing the survival rates of the IPOs in a single country cannot be generalized internationally. This study shows firm characteristics, level of market development and governance parameters that are common determinants of the IPO survival and those which are unique to the region of the IPO firm.

2.2 Hypothesis development

Previous studies have widely documented that the survivals of the newly listed firms are influenced by the firm characteristics (Hensler et al., 1997, Jain and Kini, 1999, 2000, Espenlaub et al., 2012). However, a little is known about the impact of market conditions and governance factors on the survival rates in different regions. Companies not only fail because of poor management, but also due to poor governance. For instance, failure of WorldCom, Satyam, Enron, Lehman Brothers and Bear Stearns are partly attributable to poor governance, which failed to protect shareholder's interests. Jin and Myers (2006) document that insider's find it challenging to exploit the outsiders when disclosure is transparent. This highlights the fact that good governance adds value to the firm and mitigates information asymmetry between insiders (managers) and outsiders (investors). We test whether or not the survival rates are influenced by governance-related factors:

H1: *All being equal, the survival rates of the newly listed firms are influenced by the governance issues regardless of the country or region of listing.*

Investors benefit when the stock markets are related in terms of survivability of the listing firms, for risk hedging and international diversification purposes. Understanding the relationships between the survival rates and their determinants in different markets are important for investors, in particular, long-term investors. Typically, there are similarities and differences in the listing rules and based on the region in which the firm is seeking listing. For instance, NASDAQ requires a minimum of $160 million in market capitalisation at the time of listing, while London Stock Exchange requires a minimum of £700,000. This difference might influence the survival rates of the listing firms in a specific region or might have zero impact. To explore differential listing rules might have on the survival rates; we test the following hypothesis:

H2: *The survival rates of the newly listed firms are comparable across different markets regardless of the variation in the listing rules.*

3. Data description and methodology

3.1 Sample construction and control variables

Our initial sample comprises of all IPOs listed between January 2000, and December 2008 sourced from the Thomson Financials Securities Data Companies (SDC) Platinum New Issues database. We follow LaPorta et al. (1998) to identify the list of countries to be included in our sample of international IPOs. In vein with Hasan et al. (2011), we exclude closed-end funds, right offerings, unit offerings, and IPO companies with offering price less than $5. Next we collect the firm-level IPO-date financial data for these newly listed IPO firms from SDC Platinum New Issue database, Worldscope and Datastream. To be included in the sample, we require both accounting data (total assets, earnings, sales, debt level, and cash holding) and market data (offer price, first-day closing price and market capitalization at the time of listing) to be available for the newly listed firms. We track each IPO firm in our sample until March 2015 to determine whether the firm is listed or delisted from the stock market. This information is important in estimating the survival rates of the IPO firms and their determinant's post-listing. After imposing these set of restrictions, we are left with the final sample of 7,632 IPOs, listed across thirty-two countries from 2000 to 2008.[4] Since region's governance structure plays the key role in the survival rates, we include governance parameters for the region in which the firm is listed.

LaPorta et al. (1998) established that ownership structure varies considerably around the world. Managers have incentives to divert corporate resources to themselves at the expense of minority shareholders. Recognizing the agency cost arising from conflicts of interest between the minority- and controlling-shareholders, we control for ownership concentration (sourced from Worldscope) defined as the percentage of common shares among shareholders with more than five-percent stake in the firm (Boulton et al., 2010). Following Hensler et al. (1997) and Espenlaub et al. (2012), we use age of the IPO firms measured as natural logarithm of time in years between founding / incorporation date and listing date.[5] To eliminate the impact of outliers, we trim firm-level variables, namely the initial return, and earnings and sales to total assets at the upper and lower one-percent level.

[4] Following Boulton et al. (2010), we only include countries with at least 5 or more IPOs after imposing these restrictions.
[5] We obtained the firm-level founding / incorporation dates from Worldscope and Amadeus Osiris database.

We winsorize size, cash holding, debt level, market to book and ownership concentration only at the upper one-percent level since they have a lower bound value of zero.

Besides, we include measures of economic and stock market development. In line with Lin et al. (2013), we use GDP per capita growth and stock market traded value and turnover ratio to control for the level of economic development, stock market size and liquidity. The data on country-level economic and market development factors is collected from World Bank's World Development Indicators database.[6]

Since Jackowicz et al. (2013) document the importance of governance and political influence context of the banking systems, we further investigate whether cross-country differences in governance explains the differences in the survival rates of the IPO firms. Like Black (2001), we believe that firms listed in stock markets with strong governance level, where the willingness of the regulators to forfeit the expropriation of the firm's assets by the managers will lead to a higher delisting, and hence a lower IPO survival rate. To measure the governance level internationally in our sample, we consider three types of country-level governance parameters – bondholder (or creditor) rights, anti-director minority shareholder rights and accounting disclosure.

Considering the bondholders' economic interest in the firm and their preferential status in terms of claim over the shareholders in the case of firm's liquidation, we study the effect of country-specific bondholder rights on the survival rates of newly listed firms. To control for bondholder rights, we obtain the creditor rights index from Djankov et al. (2007). The index value ranges from 0 (weak creditor rights) to 4 (strong creditor rights). The vested financial interest of the minority-shareholders and their fear regarding the expropriation of the firm's financial assets by the management or excessive financial slack at the managerial disposal may lead to long-term negative equity investments. Therefore, we next study the role

[6] For detailed definitions of the control and explanatory variables, please see Appendix A1.

of the anti-director minority shareholder rights proposed by Djankov et al. (2008) where the index value ranges from 0 (weak shareholder rights) to 6 (strong shareholder rights).[7]

Next, we control for the effect of country-level accounting disclosure quality on the survival rates. We obtain the accounting disclosure parameter from Bhattacharya et al. (2003).[8] These disclosure scores are constructed by the Center for International Financial Analysis and Research (1995) using financial variables included in the annual reports and commonly used by researchers to capture a variety of country-level characteristics. The set of disclosures include specific items from the income statement, balance sheet, and funds flow statement, as well as items related to accounting standards.

3.2 Methodology

We define survivors as stocks recently listed and continue to trade on the stock market. By default, non-survivors are stocks that are delisted from the stock market for any reason other than moving to other stock market or taken over by another firm. The survival rates of IPOs are estimated non-parametrically using the Kaplan-Meier method based on the following expression:

$$S(t_j) = \prod_{i=1}^{j} \left(\frac{n_i - d_i}{n_i} \right)^{\delta_i} \tag{1}$$

or equivalently

$$S(t_j) = \left(\frac{n_j - d_j}{n_j} \right) S(t_{j-1}) \tag{2}$$

where $S(t_j)$ is the estimated survival function in year t_j measuring the probability of survival beyond t_j conditional on the IPO j being listed at least until year t_j, n_i is the number of the IPOs that are listed and participate in the study at the start of year t_j also known as the risk set

[7] Please refer Djankov et al. (2007) for the detailed construction of creditor rights index and Djankov et al. (2008) for the detailed construction of anti-director minority shareholder rights index.
[8] Due to the unavailability of the accounting disclosure parameter for China and Russian Federation, the survival analysis using accounting disclosure as the governance factor includes only 6,818 IPOs across 30 countries, and is therefore limited to the robustness check section of the paper in table 7.

at t_j, and d_j is the number of the IPOs delisted during year t_j, δ_t is equal to one if there is a failure and zero otherwise. Alternatively, *EQ.1* can be restated as *EQ.2* to express the survival function in year t_j as the probability of survival in year t_j is conditional on the stock being listed at least up to year t_j times the survival function of the previous year t_{j-1}. We apply *EQ.2* to estimate the survival rates for each post-IPO years for both the full and regional sample.

Next, we estimate a survival model known as Accelerated Failure Time (*henceforth* AFT) model to examine the determinants of the survival rates. The AFT method allows the impact of the independent variables on survival time to vary over the post-IPO period depending on the length of time since listing. It also corrects for the fact that companies are tracked over different time periods. For instance, firms listed at the beginning of the sample period are treated similarly in the estimation model as firms listed at the end of the sample period. In the AFT model, $\exp\sum\beta_i X_i$ is an *acceleration factor*. The effect of a covariate is to extend or shrink the length of survival time by a constant relative amount $\exp\sum\beta_i X_i$. If $\exp\sum\beta_i X_i > 1$ survival time is increased, and if $\exp\sum\beta_i X_i < 1$ survival time is decreased (Bradburn et al., 2003). AFT model allows for the possibility that the determinants of the survival may be particularly pronounced in the period soon after the IPO and less so in the longer term. AFT model is typically expressed in terms of a log-linear function with respect to time (*e.g.* Hensler et al., 1997; Bradburn et al., 2003):

$$\mathrm{Ln}(\mathrm{T_j}) = \beta_0 + \beta_1 X_1 + \beta_\mathrm{p} X_p + \varepsilon_\mathrm{j} \qquad (3)$$

where $\mathrm{Ln}(\mathrm{T_j})$ is the natural logarithm of the survival time or time to failure. As AFT is a parametric model, it is necessary to specify the distribution of the baseline survival function. We use the likelihood ratio or Wald test to determine the appropriate distribution in case of nested models, such as comparing the Weibull against the exponential distribution, or the gamma against the Weibull or log-normal distributions. The AIC is the appropriate test to choose the best-fitting model in the case of a non-nested model between the log-logistic and the log-normal distribution. The AIC is defined as:

$$AIC = -2LnL + 2(k + c) \tag{4}$$

where L is the maximized value of the likelihood function, k is the number of model covariates and c is the number of model-specific distributional parameters. Either of the log-normal or log-logistic models has two distributional parameters *i.e.* $c=2$. The AIC test shows that log-normal distribution has lower AIC value than the log-logistic model, and hence we select the log-normal distribution.

For robustness check and comparative purposes, we also estimate the Cox Proportional Hazard model as applied by Carpentier and Suret (2011). The Cox model makes no assumption about the distribution failure. The dependent variable in the Cox model measures the risk of failure compared to the survival time in the AFT model. In the Cox model, the marginal effect of an independent variable is measured by the *so-called* hazard ratio. A positive coefficient implies a hazard ratio (calculated as the exponential coefficient from the Cox model) of greater than one, suggesting an increase in the covariate increases the failure rate. While, a negative coefficient implies a hazard ratio of less than one, indicating that an increase in the covariate reduces the failure rate (Kleinbaum, 1996).

4. Empirical findings

4.1 Descriptive statistics

Table 1 presents the summary statistics on the IPO firms' listing-time characteristics, overall economic conditions of the domestic market at the time of listing, and general stakeholder protection and disclosure level of the market where the firm decides to list. Empirics are discussed for the full-sample of thirty-two countries, and across four sub-regions for the IPOs listed from 2000 to 2008.

Panel A of table 1, provides the number of survived and failed IPOs for the full-sample and for the four sub-regions tracked up to the end of March 2015. Expectedly, a significant number of IPOs hit the market at the peak of the dotcom bubble around the turn of the century, not only in the North-America, but globally *i.e.* Europe, BRICS and East-Asia & Australia. The subsequent market crash had a considerable negative impact on the global IPO market whereby the overall listing numbers dropped to half for the subsequent years (at least until 2004), followed by the gradual revival of the IPO market. We observe a gradual but steady increase in the number of IPOs from 2002 to 2007. With increased listings, the survival rate of the IPO firms remained stagnant for the remaining sample period, with European IPOs experiencing a significantly higher failure rate – in the range of 30 to 40 percent. Lastly, in 2008 with the onset of the sub-prime crisis and global recession, there was a sudden drop in the number of listings internationally. This was primarily driven by the significant roll-back of the pre-planned IPOs by their respective underwriters and investment banks on the pretext of under-subscription and under-pricing.

The summary statistics of the main variables across thirty-two countries are reported in panel B. Besides the standard firm-specific controls used in conventional IPO literature, we also include firm-specific proxies to control for agency aspect, level of market development and stakeholder protection. The mean (median) value of the first-day return is 33.1 percent (12.2 percent) which is consistent with the extant literature (Boulton et al. 2010; Lin et al. 2013). Furthermore, expectedly most IPOs, have limited liquid assets (*i.e.* cash holding) with a moderate level of debt on their books, and concentrated ownership. The empirical distribution of our variables; as shown by 5^{th} and 95^{th} percentile values is reasonable and does not suggest a selection bias. Given the importance of governance variables, we include various governance characteristics and test the effect of the investor rights on the survival rate using a number of different yardsticks. Our governance parameters show significant variation from 1 to 4 (bondholder rights), 1 to 5 (shareholder rights) and 58 to 85 (accounting disclosure). Similarly, reasonable magnitudes apply to the overall level of market development where the IPO firm decides to list. Lastly, after closely tracking 7,632 IPO listings, we find about 22% (*i.e.* 1,687 delistings) of the IPOs listed during our sample period fail to survive.[9]

In panel C of table 1, we report the descriptive statistics discussed in panel B, segregated by survived and failed IPO firms for the full sample of thirty-two countries and four sub-regions. We observe a significant number of delistings of IPO firms in regions with improved governance, in particular, North-America and Europe, compared to elsewhere. This could be due to the stringent listing and governance rules, which prevent the expropriation of firm finances and shareholder investments at the hands of the managers (Black, 2001). Besides the governance issues, failed firms in all regions are young with lower first-day return, significantly lower market capitalization, profitability, sales, growth opportunities, and cash holding than the surviving firms. In terms of leverage, failed firms have a relatively higher leverage ratio than firms that have survived. Nevertheless, survived firms for which insider ownership is concentrated, exhibit clear signs of principal–agent conflict in the management of the firm. Finally, firms listed in high-growth markets (GDP per capita growth) with a liquid (turnover ratio) stock exchange tend to exhibit higher survival rates.

[Please insert table 1 about here]

4.2 Survival rates using Kaplan-Meir process

The non-parametric survival rate for each region is estimated using Kaplan-Meier method and presented in table 2. The table reports the survival rates over three- and five-year post-listing for the full sample, followed by the four sub-regions, *i.e.* North-America, Europe, BRICS, and East-Asia & Australia. The survival rate for the full sample ranges between 87

[9] In vein with popular IPO survival literature (*e.g.* Hensler et al. 1997; Jain and Kini, 1999, 2000; Espenlaub et al. 2012; *etc.*), we too observe *circa* 2% of the firms in our sample delisted due to liquidation and bankruptcy.

percent (for the IPO firms listed in 2004) and 91 percent (IPOs listed in 2003) over three-years post-listing and between 69 percent (IPOs in 2000) and 88 percent (IPOs in 2008) over five-years. Provided that the full sample includes IPOs across thirty-two markets from all the regions, it is important to identify a sub-region with the highest survival rate. To understand variation in the survival rates at the regional level, we estimate the survival rates by region of the IPO listing. On an average, IPOs listed in North-America have moderate survival rates over the three-year period (at *circa* 86.5 percent), while over five-years, post-listing the survival rates declines to as low as 62 percent, which is equivalent to a 38 percent failure rate. This figure is comparable to the failure rates reported for the North-American IPOs (Hensler et al., 1997; Jain and Kini, 2000). Moving to the European IPOs, the results show that the survival rates over three-years are 71 percent (for listings in 2003 and 2004) and decreases to further 49 percent (for the IPOs in 2003) over a five-year period post-listing. IPOs listed in the five BRICS countries exhibit a relatively higher survival rate, ranging from 90.5 percent (IPOs in 2006) to 93.5 percent (IPOs in 2001) for three-years post-listing and from 74 percent (IPOs in 2000) percent to 91 percent (IPOs in 2008) for five-years post-listing. In terms of failure, these figures suggest a mean failure rate of 8 and 18.5 percent respectively. Lastly, East-Asian & Australian IPOs have the highest survival rates over three- and five-years post–listing compared to all the sub-regions*, i.e.* an average survival rate of about 96.8 percent over three-years and 93.2 percent over a five-year period.

[Please insert table 2 about here]

Overall, the results of table 2 show that the IPOs listed in the East-Asia & Australia have the highest survival rates, followed by those in the BRICS economies, while those listed in Europe exhibit the worst short- and long-term performance. These results establish the first premises of our empirical findings, suggesting that region where the IPO firm is listed influences its survival post-listing. Arguably, differences in the IPO features, governance and level of market development could explain the differences in the survival rates across different regions. To address this concern, we use multivariate analysis (AFT model) controlling for the IPO firm characteristics, level of market development and country-level stakeholder protection where the firm is listed.

4.3 Multivariate analysis

4.3.1 Survival and IPO firm characteristics

Table 3 reports the base regression results for the AFT model. We examine the determinants of the IPO survival rates using the firm-level characteristics of the IPO firms at the time of listing. We report the results for the full sample of 7,632 IPO firms, followed by the four sub-regions. For the full sample, we find that conventional listing-time IPO firm-level characteristics play a significant role in determining their survival rate. Expectedly, on one hand the initial returns, size of the firm, earnings and sales accelerate the survival time of the new listings, while IPO mispricing and the level of debt decelerates the survival time post-listing. Consistent with the related prior studies (Hensler et al., 1997, Jain and Kini, 2000 and Espenlaub et al., 2012), we note that the newly listed firms, characterised by high initial returns at the time of listing, size, profitability and sales are likely to survive longer than their counterpart. In terms of leverage, highly levered firms at the time of listing tend to have lower survival rates, but the evidence is not significant at any conventional level. Consistent with Ahmed and Jelic (2014) for the U.K. IPOs, highly leveraged IPOs are less likely to attract long-term investors at the time of listing and the prospect of raising capital is rather uncertain.

However, since different markets attract different types of IPOs, it is not realistic to assume that same firm-level features are equally important across all the regions. Therefore, to address this concern, we investigate the survival rates for the IPOs in each of the four sub-regions individually. Our results show that profitability, both in terms of earnings and sales are distinctly important determinant of the survival for the IPOs listed in North America. The effect of other factors is theoretically consistent, but statistically insignificant. In Europe, initial returns, size and earnings accelerate the survival rates post-listing. Among the BRICS countries and in the East-Asia & Australian region, size of the IPO firms and earnings are important determinants. However, in BRICS region high IPO mispricing and in the East-Asia & Australia, high leverage plays a detrimental role in the new list survival rates. In a nutshell, our results suggest that firm characteristics at the time of listing are important, but there is a significant variation in what determines the survival and where at the time of listing. Nonetheless, profitability of the IPO firm at the time of listing is an important determinant of the survival rates across all the regions. This suggests that regardless of the market in which the company is listed, earning is a global factor that accelerates the survival time.

[Please insert table 3 about here]

4.3.2 Survival and level of market development and stakeholder protection

To investigate the impact of market developments and regional investor protection level, we estimate the survival rates using the level of market development and governance parameters. Panel A of table 4 presents the results for the impact of level of market development, while panel B for the stakeholder protection on the IPO firm survival rate. For the full sample, we find that GDP per capita growth and stock market liquidity (*i.e.* turnover ratio) and size (StTraVal) play a significant role in determining the IPO firm survival rate. Increase in the overall growth opportunities in the economy (*i.e.* GDP per capita growth) by one percent increases the survival time up to 10.9 percent compared to 0.4 percent and 0.2 percent for every percentage point increase in the stock exchange liquidity and size respectively. While exploring the variations in the level of market development across different regions, we find that the level of stock exchange liquidity where the IPO firm is listed plays a crucial role in the survival post-listing. In brief, high market liquidity and size of the stock exchange accelerate the survival time of firms listed in North-America and Europe. However, for firms listed in BRICS and East-Asia & Australia, the effect of market liquidity dominates that of GDP per capita growth or size of the stock market.

Panel B of table 4 reports the results on the significance of the stakeholder protection *viz.* bondholder rights and anti-director minority shareholder rights in estimating the new firm survival rate. For the full sample, we find that strong bondholder rights decelerate the survival time post-listing, while strong shareholder's rights accelerate the survival time. The evidence is also statistically significant and consistent across all the four sub-regions except the North-America where only the shareholder rights play a deciding factor. In summary, country-level stakeholder (both creditors and shareholders) rights influence the new list survival rate globally by establishing the balance of power between debt- and equity-claimants, with shareholders pulling significantly more weight in dictating the fate of the newly listed IPO firm. Generally, shareholders in markets with strong anti-director rights prefer to keep the new IPO firm afloat, due to their vested financial interest in the firm. Together, our results show that the IPO firms listed in regions with high stock market liquidity, and strong shareholder rights are likely to survive longer post-listing.

[Please insert table 4 about here]

Table 5 reports the factors of the survival rates using AFT model for the IPO firm characteristics and level of market development (panel A), and the IPO firm characteristics and level of investor protection (panel B). Here, we aim to understand the interaction of IPO-date firm-level variables with market development and country-level governance mechanism in deciphering the survival rates. In panel A of table 5, we observe that the IPO firm characteristics and the level of market development proxies play significant role in deciding the likelihood of survival of the newly public firm. The results are consistent with those reported earlier for table 3 and table 4 (panel A) separately. To summarize, firms listed in countries with relatively liquid stock exchanges tend to survive longer. In the U.S. and Canada, size and profitability are important in terms of firm-level characteristics, while market liquidity and size of the stock market are also equally important for the long-term fate of the newly listed firm. In Europe, other than initial return, the survival rate of the IPO firms is determined by similar characteristics. Among BRICS and East-Asia & Australia size of the IPO, profitability, and market turnover increase the survival rates. Besides, higher domestic GDP per capita growth and lower listing-time debt level of the firm increase the survival rates among the nine East-Asian & Australian economies.

Table 5, panel B shows qualitatively similar results, when we study the determinants of the survival rates using firm characteristics and level of investor protection. For the full sample, we find that both firm characteristics and governance variables jointly determine the survival rate at the time of listing. Strong minority shareholder rights accelerate the survival time, while strong creditor rights decelerate the survival post-listing. Also the results show that firms with significant mispricing at the time of listing and high leverage are likely to fail post-listing. However, regional analysis shows that size, earnings and strong shareholder rights are decisive factors of the survival rates globally. It is clear from the results that the IPO firm survival is principally determined by competing creditor and minority shareholder rights. However, managers are likely to pursue equity-holders interest since both parties have a vested incentive to minimize the agency cost of equity.

[Please insert table 5 about here]

In table 6, we pool all the listing-time IPO firm-level features, level of market development and investor proxies in one single model. The likelihood ratio statistics show the model is significant at 1 percent level for the full sample and across four sub-regions. For the full sample, we find that profitability improves the survival time, while IPO mispricing and debt-level reduces the survival rates. The evidence is statistically significant at a conventional

5 percent level. Consistent with Lin et al. (2013), all three market development parameters (*i.e.* GDP per capita growth, and domestic stock exchange liquidity and size) accelerate the survival time. This is in line with LaPorta et al. (2006) study, which documents that countries with significant growth opportunities and considerably bigger and liquid equity market provide an appropriate platform for the newly public firms to survive. For the regional analysis, we find that size, earnings, turnover and anti-director minority shareholder rights are key determinants of the survival rates for all the regions. Except for Europe, trading volume also increases the survival rates' post-IPO, while creditor rights decrease the survival rates for all regions except in the North-America, where the effect is statistically insignificant.

Together, our results show that size of the IPO, firm-level profitability, market liquidity and strong shareholder rights help in discerning the survival rates globally. Nonetheless, profitable firms, which decide to list in economically and financially developed stock exchanges with a higher turnover ratio are more likely to survive in the long run. Furthermore, our findings indicate that creditors exert significant and far-reaching influence over new firm survival decision-making. Nonetheless, the importance of the agency cost of equity predominates not only globally but also at the regional level.

[Please insert table 6 about here]

5. Robustness check

In this section, we perform a battery of sensitivity analysis to ensure the robustness of our results. To further address the problem of potentially omitted variables, herein we include an array of listing-time firm-level and country-level control variables to test the robustness of the empirical findings established earlier in this paper. In table 7, following conventional IPO literature, we test the impact of popular IPO-date firm level characteristics commonly used to address the agency-cost based lifecycle issues and level of disclosure required by the firms once they start trading on a public platform.[10] In model I of table 7, we find that firms that avoid the haste to list on the stock market, and rather concentrate on building a reputable business model, are more likely to survive in the long run, *i.e.* mature firms tend to increase the time-to-failure with an increase in age. Hence if the firm is more established at the date of IPO, the expected survival rate is higher, which is consistent with Espenlaub et. al (2012).[11] Next, in line with Hensler et al. (1997), analysis of the insider ownership concentration (model II, table 7) shows that in case of concentrated ownership, the survival time is longer. Although, this finding is contradictory to the rationale put forward by LaPorta et al. (1998) and represents entrenchment, the possible explanation is that post-listing, insiders / managers with higher financial interest vested in the firm might work towards the overall betterment of the newly public firm. In model III of table 7, we note that the IPO firms with relatively higher level of cash and short-term investments are more likely to hedge against unpredictable adverse future events and thus survive longer. Finally, in a model IV of table 7, we observe that the firms that list in markets with stricter mandatory accounting disclosure requirements face tougher time in surviving post-IPO. This is in the vein with Black (2001), *i.e.* the regulators in markets with meaningful listing standards are not only willing to enforce the rules by fining, but also delist companies that failed to protect the interest of the stakeholders in the firm. Our results are robust across all the four sub-regions.

To further examine the robustness of our results in the absence of distributional assumption, we use Cox Proportional Hazard model (see Appendix A2). In the Cox model, the survival rates can take any shape: increasing, decreasing or monotonic. Here, a positive co-efficient in an explanatory variable indicates that an increase in the variables, is associated

[10] For brevity, in table 7 we only present the results for the full sample of thirty-two countries for the four new IPO survival explanatory variables. For the new explanatory variables, across the four sub-regions, the results are statistically significant at 10 percent or higher level with theoretically consistent sign as reported for the full sample in table 6 and 7. The results for the four sub-regions are available from the authors upon request.

[11] For the ease of illustration, we mainly focus on the four new explanatory variables used in table 7 since the results for the regular listing-time IPO firm characteristics, level of market development and investor protection are consistent with those reported in table 6 with respect to the regular set of control / explanatory variables.

with an increase in the hazard and therefore, should experience lower duration until graduation *viz.* delisting. We find the results for the full sample and for the four distinct sub-regions to be theoretically consistent with the results reported in table 6.

Ultimately, the IPO literature distinguishes between the delisting of the firm caused by the listed firm's actual failure and its merger and / or acquisition as two different outcomes (Howton, 2006). Taking this into consideration, we check if the results hold by excluding all firms that have been delisted due to merger and / or acquisition post-listing. We also exclude all cross-listed firms and firms with ADRs listed outside their domestic market. Appendix A3 presents the IPO firm-level results for the full sample of 5,744 IPO firms and for the four sub-regions. The results are consistent across all the models, and the coefficients of the control variables remain qualitatively similar to those reported and discussed in table 6.

[Please insert table 7 about here]

Collectively, the results show that IPO firm characteristics, regulatory and governance indicators and level of domestic market development parameters have a significant impact on the survival rates of the newly listed firms in short and long-term horizon. In general, mature profitable firms of reasonable size and listed in economies with high-growth opportunities, and liquid stock markets have better odds of surviving. Lastly, different governance and disclosure level have a significant impact in their own right on the likelihood of surviving.

6. Conclusion

This study examines the determinants of the survival rates using firm characteristics, financial and economic level of market development and quality of investor protection at the time of listing. We classify IPO firms listed in thirty-two markets into four regions – North-America, Europe, BRICS, and East-Asia & Australia. Our sample consists of 7.632 IPO firms listed in these four regions between 2000 and 2008, with their listing status being tracked up to the end of March 2015.

We find; across all regions, the survival rates over three-year post-listing are comparable, but differ substantially over a five-year post-listing period. Based on the listing year and region, firms listed in Europe have the lowest survival rates, while those listed in East-Asia & Australia depict the highest survival rates' post-listing. We use AFT and Cox model to examine the effect of differences in the IPO characteristics, extent of investor rights, and the level of market development across all the regions. Indeed, we find that the impact of listing-time firm-level attributes, market development and governance indicators are different across the regions. Nonetheless, earnings, stock market liquidity and anti-director minority shareholder rights at the time of listing determine the survival rates globally, while the role of other firms' characteristic, market development and governance factors varies by region of the IPO firm. Our results show a significant positive influence of the age, level of liquid assets and ownership concentration in explaining the long-term fate of the firm. Our results are robust to a number of alternative model specifications, standard error clustering, and sample decompositions.

This study contributes to the IPO survival literature by showing IPO characteristics that determine the survival rates internationally. Further, we show that the economic and financial level of market development and domestic governance and regulatory indicators play a vital role in accelerating the survival of the IPOs post-listing. We anticipate that these results are useful for the policymakers in understanding the determinants of new-list survival over long-run. Knowing which listing time factor is more likely to accelerate or decelerate the survival is critical for the international investors when deciding whether they want to participate in an IPO beyond the realms of their domestic market. While extant literature focuses on a single-market, this paper concentrates on what explains the survival of the IPOs globally and across different regions.

Reference

Ahmed, W., Jelic, R., 2014. Lockup agreements and survival of UK IPOs. Journal of Business Finance & Accounting 41; 717-742.

Baele, I., Ferrando, A., Hordahl, P., Krylova, E., Monnet, C., 2004. Measuring European financial integrations. Oxford Review of Economic Policy 20; 509-530.

Bhattacharya, U., Daouk, H., Welker, M., 2003. The world price of earnings opacity. The Accounting Review 78; 641–678.

Black, B.S., 2001. The legal and institutional preconditions for strong securities markets. UCLA Law Review 48; 781-855.

Boulton, T.J., Smart, S.B., Zutter, C.J., 2010. IPO underpricing and international corporate governance. Journal of International Business Studies 41; 206-222.

Bradburn, M., Clark, T., Love, S., Altman, D., 2003. Survival analysis part II: Multivariate data analysis–an introduction to concepts and methods, British Journal of Cancer 89; 431-436.

Carpentier, C., Suret, J., 2011. The survival and success of penny stock IPOs: Canadian evidence. Small Business Economics 36; 101-121.

Craig, S., Putka, G., 2002. Robertson Stephen rode wave of IPOs, now fights for survival. Wall Street Journal; June 13.

Djankov, S., McLiesh, C., Shleifer, A., 2007. Private credit in 129 countries. Journal of Financial Economics 84; 299–329.

Djankov, S., La Porta, R., Lopez-de-Silanes, F., Shleifer, A., 2008. The law and economics of self-dealing. Journal of Financial Economics 88; 430–465.

Elyasiani, E., Mester, L.J., Pagano, M.S., 2014. Large capital infusions, investor reactions, and the return and risk-performance of financial institution over the business cycle. Journal of Financial Stability 11; 62-81.

Erb, C.B., Harvey, C.R., Viskanta, T.E., 1994. Forecasting international equity correlations. Financial Analyst Journal 50; 32-45.

Espenlaub, S., Khurshed, A., Mohamed, A. 2012. IPO survival in a reputational market. Journal of Business Finance and Accounting 39; 427-463.

Fama, F., French, K., 2004. New lists: Fundamentals and survival rates. Journal of Financial Economics 73; 229-269.

Friedman, J., Shachmurove, Y., 1997. Comovements of major European community stock markets: A vector autoregression analysis. Global Finance Journal 8; 257-277.

Hasan, I., Kobeissi, N., Wang, H., 2011. Global equity offerings, corporate valuation and subsequent international diversification. Strategic Management Journal 32; 787-796.

Hensler, D., Rutherford, R., Springer, T., 1997. The survival of initial public offerings in the aftermarket. Journal of Financial Research 20; 93-110.

Howton, S.W., 2006. Effect of governance characteristics on the state of the firm after an initial public offering. Financial Review 41; 419-433.

Huyghebaert, N., Quan, Q., 2006. Interdependence between European, US and Japanese stock markets: Did the Euro promote further integration? International Economics and Finance Journal 1; 247-274.

Huyghebaert, N., Wang, L. 2010. The co-movement of stock markets in East Asia did the 1997-1998 Asian financial crisis really strengthen stock market integrations? China Economic Review 21; 98-112.

Jackowicz, K., Kowaleswski, O., Kozlowski, L., 2013. The influence of political factors on commercial banks in Central European countries. Journal of Financial Stability 9; 759-777.

Jain, B., Kini, O., 1999. The life cycle of initial public offerings. Journal of Business Finance and Accounting 26; 1281-1307.

Jain, B., Kini, O., 2000. Does the presence of venture capitalists improve the survival profile of IPO firms? Journal of Business Finance and Accounting 27; 1139-1176.

Jain, B., Martin, C., 2005. The association between audit quality and post-IPO performance: A survival analysis approach. Review of Accounting and Finance 4; 50–75.

Jin, L., Myers, S.C. (2006). R^2 around the world: New theory and new tests. Journal of Financial Economics 79, 257–292.

Kaufman, G.G., 2014. Too big to fail in banking: What does it mean? Journal of Financial Stability 11; 62-81.

Karolyi, A., Stulz, M., 1996. Why do markets move together? An investigation of US-Japan stock returns co movements. Journal of Finance 51; 951-986.

Kleinbaum, D., 1996. Survival analysis: A self-learning text Springer Verlag: New York.

Kooli, M., Meknassi, S., 2007. The survival profile of U.S. IPO issuers: 1985-2005. Journal of Wealth Management 10; 105-119.

LaPorta, R., Lopez-de-Silanes, F., Shleifer, A., Vishny, R., 1998. Law and finance. Journal of Political Economy 106; 1113–1155.

LaPorta, R., Lopez-de-Silanes, F., Shleifer, A., 2006. What works in securities laws? Journal of Finance 61; 1–32.

Lin, H.L., Pukthuanthong, K., Walker, T.J., 2013. An international look at the lawsuit avoidance hypothesis of IPO underpricing. Journal of Corporate Finance 19; 56-77.

Weild, K., Kim, E., Newport, L., 2013. Making stock markets work to support economic growth: Implications for governments, regulators, stock exchanges, corporate issuers, and their investors. OECD Corporate Governance Working Paper Series No. 10, OECD Publishing.

Table 1: Descriptive statistics of the variables used in this study.

This table reports the summary statistics for the variables used in this study for the newly listed IPOs across thirty-two countries from 2000 to 2008. Panel A shows the annual distribution of survived and failed IPO firms whose listing status is tracked up to March 2015 across full sample of thirty-two countries and four sub-regions used in this study. Panel B presents the descriptive statistics of all the variables used in this study. N refers to the number of observations, Mean is the average, Std. dev. is the standard deviation, Median is the median, 5%, Q1, Q3 and 95% are the 5^{th}, 25^{th}, 75^{th} and 95^{th} percentile values. Panel C shows the average values of the variables for the survived and failed IPO firms across full sample of thirty-two countries and four sub-regions used in this study. For the detailed definition and construction of the proxy variables and distribution of thirty-two countries across four sub-regions, please refer to the Appendix A1.

Panel A: Distribution of the survived and failed IPO firms listed between 2000 and 2008.

IPO Years	Full Sample			North-America			Europe			BRICS			East-Asia & Aust.		
	Survived (#)	Failed (#)	Failed (%)	Survived (#)	Failed (#)	Failed (%)	Survived (#)	Failed (#)	Failed (%)	Survived (#)	Failed (#)	Failed (%)	Survived (#)	Failed (#)	Failed (%)
2000	662	295	30.83	93	21	18.42	191	169	46.94	109	11	9.17	269	94	25.90
2001	394	169	30.02	42	21	33.33	79	80	50.31	42	3	6.67	231	65	21.96
2002	394	143	26.63	44	26	37.14	49	38	43.68	59	6	9.23	242	73	23.17
2003	464	123	20.95	46	20	30.30	27	31	53.45	71	2	2.74	320	70	17.95
2004	766	248	24.46	106	49	31.61	110	77	41.18	125	7	5.30	425	115	21.30
2005	766	236	23.55	110	55	33.33	121	71	36.98	105	11	9.48	430	99	18.71
2006	945	223	19.09	127	36	22.09	200	101	33.55	223	13	5.51	395	73	15.60
2007	1121	194	14.75	186	32	14.68	225	73	24.50	299	34	10.21	411	55	11.80
2008	433	56	11.45	73	20	21.51	43	12	21.82	121	4	3.20	196	20	9.26
Total	5945	1687	22.10	827	280	25.29	1045	652	38.42	1154	91	7.31	2919	664	18.53

Panel B: Descriptive statistics for the variables, across thirty-two countries for all the IPO firms listed between 2000 and 2008.

Variables	N	Mean	Std. dev.	5%	Q1	Median	Q3	95%
Day one ret. (%)	7632	33.098	68.755	-26.039	0.219	12.200	41.760	165.597
Mkt. Cap. (US$ Mil.)	7632	606.957	4449.336	5.136	23.568	71.520	259.058	1857.175
Total Assets (US$ Mil.)	7632	854.928	12555.523	6.400	9.308	30.875	102.475	1147.034
EBIT to TA (%)	7632	5.184	19.653	-40.000	-0.280	6.634	13.527	33.087
Sales to TA (%)	7632	88.286	86.520	0.587	16.691	70.595	127.894	262.573
MTB	7632	3.496	4.540	0.640	1.260	2.190	3.850	10.390
Debt to TA (%)	7632	23.555	50.018	0.000	0.258	14.625	35.593	67.506
Growth_TA (%)	7632	43.439	81.440	-24.620	5.770	24.602	56.909	176.107
Age (Years)	6396	13.366	17.059	0.000	2.000	8.000	17.000	49.000
Insider own (%)	4495	53.667	23.958	6.727	37.280	57.660	71.630	88.632
Cash to TA (%)	6213	40.381	33.066	1.256	11.670	30.600	67.860	99.210
GDPpcG (%)	7632	3.936	3.429	0.098	1.719	2.527	5.365	12.072
Turnover (%)	7632	110.541	58.228	34.770	71.874	101.404	138.661	216.458
StTraVal (%)	7632	112.921	81.538	27.391	56.675	89.412	149.144	283.770
Bondholder right (0 to 4)	7632	2.253	1.012	1.000	2.000	2.000	3.000	4.000
Shareholder right (0 to 6)	7632	3.561	1.183	1.000	3.000	3.500	5.000	5.000
Acc. Disc.	6818	73.081	7.875	58.000	68.000	75.000	79.000	85.000
Survival	7632	0.779	0.415	0.000	1.000	1.000	1.000	1.000

Panel C: Average value for the variables, across thirty-two countries and four sub-regions for the survived and failed IPO firms listed between 2000 and 2008.

	Full Sample		North-America		Europe		BRICS		East-Asia & Aust.	
Variables	Survived	Failed	Survived	Failed	Survived	Failed	Survived	Failed	Survived	Failed
Day one ret. (%)	34.962	26.531	21.565	12.676	19.729	10.982	64.464	55.350	35.679	35.102
Mkt. Cap. (US$ Mil.)	648.386	460.962	841.159	451.945	742.175	396.133	1546.484	1175.021	430.560	205.138
EBIT to TA (%)	6.060	2.100	5.621	-0.623	1.782	-2.706	13.558	12.537	8.339	6.397
Sales to TA (%)	90.729	79.679	54.357	45.148	87.459	80.723	88.890	84.613	102.931	92.540
MTB	3.420	3.763	3.255	4.137	4.298	4.317	3.126	3.967	3.019	3.405
Debt to TA (%)	22.955	25.668	19.553	23.315	25.284	30.790	26.120	24.004	20.769	23.444
Growth_TA (%)	48.675	41.953	61.757	50.846	58.021	53.095	43.298	40.409	31.822	39.716
Age (Years)	14.046	10.755	10.071	8.231	14.374	10.529	13.884	9.478	15.227	12.180
Insider own (%)	54.711	50.414	46.173	40.141	53.043	48.958	62.329	60.103	55.534	55.812
Cash to TA (%)	41.796	40.003	47.344	57.687	40.268	38.418	36.326	29.435	41.870	40.095
GDPpcG (%)	4.253	2.820	1.565	1.668	2.416	2.495	8.778	8.722	3.883	2.816
Turnover (%)	112.794	102.601	144.708	113.867	123.879	114.703	111.284	107.259	100.380	85.328
StTraVal (%)	112.599	114.057	191.495	142.026	125.271	132.770	79.951	83.594	98.617	88.063
Bondholder right (0 to 4)	2.188	2.481	1.000	1.000	2.273	2.966	1.938	1.846	2.594	2.715
Shareholder right (0 to 6)	3.472	3.873	3.366	3.596	3.474	4.046	2.503	2.374	3.885	4.026
Acc. Disc.	72.082	76.281	75.634	75.404	75.519	79.000	60.432	59.308	71.462	74.645

Table 2: Survival rate of the IPO firms 3- and 5-years after listing.

This table reports the survival rates of the IPO firms using Kaplan-Meier method for the newly listed IPOs between Jan: 2000 and Dec: 2008, tracked from Jan: 2000 to Mar: 2015. Kaplan-Meier results are reported for the proportion of the IPO firms survived beyond 3-years (short-term) and 5-years (long-term) after listing for the full sample across thirty-two countries and four sub-regions used in this study.

IPO Years	Full Sample		North-America		Europe		BRICS		East-Asia & Aust.	
	3-Years	5-Years	3-Years	5-Years	3-Years	5-Years	3-Years	5-Years	3-Years	5-Years
2000	0.8871	0.6917	0.9737	0.6286	0.7944	0.5306	0.9146	0.7410	0.9917	0.9083
2001	0.9076	0.6998	0.8889	0.6667	0.8302	0.4969	0.9358	0.7804	0.9556	0.9333
2002	0.9065	0.7364	0.8714	0.6970	0.8118	0.5765	0.9206	0.7683	0.9846	0.9077
2003	0.9097	0.7905	0.8788	0.8158	0.7069	0.4655	0.9282	0.8205	0.9863	0.9726
2004	0.8724	0.7577	0.8117	0.6883	0.7043	0.5914	0.9184	0.7885	0.9924	0.9470
2005	0.8743	0.7645	0.8121	0.6667	0.7813	0.6302	0.9168	0.8129	0.9224	0.9052
2006	0.8767	0.8091	0.8282	0.7791	0.7973	0.6645	0.9060	0.8440	0.9534	0.9449
2007	0.9019	0.8525	0.8945	0.8532	0.8389	0.7550	0.9206	0.8820	0.9369	0.8979
2008	0.8978	0.8855	0.8280	0.7849	0.8000	0.7818	0.9120	0.9074	0.9920	0.9680

Table 3: Firm-level determinants of the IPO survival.

This table reports the baseline regression results for the IPO's listing-time firm-level characteristics. We use the Accelerated Failure Test (AFT) model to study the determinants of the survival rate of the newly listed IPOs between Jan: 2000 and Dec: 2008, tracked to Mar: 2015. AFT model results are reported for the full sample across thirty-two countries and four sub-regions used in this study. We control for the firm–level industry and year fixed effects using dummy variables. 'Observations' is the number of observations used in each regression model. For the detailed definition and construction of the proxy variables, please refer to the Appendix A1.

	Full Sample		North-America		Europe		BRICS		East-Asia & Aust.	
Variables	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.
Day one ret.	0.0015	(0.0045)	0.0026	(0.1798)	0.0022	(0.0212)	0.0019	(0.1472)	0.0004	(0.6131)
Ln_Mkt Cap.	0.0523	(0.0007)	0.0154	(0.6161)	0.1343	(0.0000)	0.1110	(0.0475)	0.0518	(0.0180)
EBIT to TA	0.0058	(0.0000)	0.0120	(0.0101)	0.0060	(0.0005)	0.0084	(0.0862)	0.0002	(0.0950)
Sales to TA	0.0008	(0.0627)	0.0023	(0.0544)	0.0003	(0.6752)	0.0011	(0.2977)	0.0010	(0.1000)
MTB	-0.0163	(0.0017)	-0.0215	(0.3028)	-0.0008	(0.8929)	-0.0313	(0.0630)	-0.0046	(0.5470)
Debt to TA	-0.0008	(0.0094)	0.0012	(0.5441)	-0.0003	(0.1625)	0.0034	(0.6719)	-0.0035	(0.0024)
Growth_TA	-0.0006	(0.1121)	0.0004	(0.7220)	-0.0003	(0.5448)	0.0015	(0.3834)	-0.0007	(0.3012)
Constant	3.3343	(0.0000)	3.3134	(0.0000)	2.4112	(0.0000)	5.8640	(0.0000)	4.0199	(0.0000)
Year FE	Yes		Yes		Yes		Yes		Yes	
Industry FE	Yes		Yes		Yes		Yes		Yes	
Likelihood	-5433		-859		-1688		-393		-2268	
Observations	7627		1106		1694		1245		3582	

Table 4: Level of market development and stakeholder protection in determining the IPO survival.

This table reports the baseline regression results for the IPO listing-time level of market development (Panel A), and the level of stakeholder protection (Panel B) for the market / stock exchange where the IPO firm decides to list. We use the AFT model to study the determinants of the survival rate of the newly listed IPOs between Jan: 2000 and Dec: 2008, tracked from Jan: 2000 to Mar: 2015. AFT model results are reported for the full sample across thirty-two countries and four sub-regions used in this study. We control for the firm–level industry and year fixed effects using dummy variables. 'Observations' is the number of observations used in each regression model. For the detailed definition and construction of the proxy variables, please refer to the Appendix A1.

	Panel A: Level of market development									
	Full Sample		North-America		Europe		BRICS		East-Asia & Aust.	
Variables	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.
GDPpcG	0.1086	(0.0000)	0.0553	(0.5357)	0.0150	(0.7152)	0.0091	(0.9103)	0.0762	(0.0000)
Turnover	0.0042	(0.0000)	0.0120	(0.0695)	0.0068	(0.0000)	0.0066	(0.0603)	0.0037	(0.0000)
StTraVal	0.0020	(0.0023)	0.0131	(0.0015)	0.0051	(0.0000)	-0.0051	(0.2030)	-0.0006	(0.1432)
Constant	3.0635	(0.0000)	3.0646	(0.0000)	2.7805	(0.0000)	4.9161	(0.0000)	3.2047	(0.0000)
Year FE	Yes		Yes		Yes		Yes		Yes	
Industry FE	Yes		Yes		Yes		Yes		Yes	
Likelihood	-5440		-874		-1721		-405		-2247	
Observations	7627		1106		1694		1245		3582	

	Panel B: Governance factors									
	Full Sample		North-America		Europe		BRICS		East-Asia & Aust.	
Variables	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.
Bondholder right	-0.1307	(0.0000)	-0.0001	(0.4125)	-0.1182	(0.0000)	-1.2046	(0.0132)	-0.1368	(0.0767)
Shareholder right	0.1967	(0.0002)	1.0048	(0.0000)	0.2569	(0.0000)	0.1269	(0.0461)	0.1699	(0.0181)
Constant	4.6421	(0.0000)	7.0547	(0.0000)	4.2245	(0.0000)	2.7035	(0.0010)	4.8374	(0.0000)
Year FE	Yes		Yes		Yes		Yes		Yes	
Industry FE	Yes		Yes		Yes		Yes		Yes	
Likelihood	-5469		-878		-1678		-402		-2268	
Observations	7627		1106		1694		1245		3582	

Table 5: This table reports the baseline regression results for the IPO listing-time firm-level characteristics and the level of market development (Panel A), and firm-level characteristics and the level of stakeholder protection (Panel B) of the market / stock exchange where the IPO firm decides to list. We use the AFT model to study the determinants of the survival rate of the newly listed IPOs between Jan: 2000 and Dec: 2008, tracked from Jan: 2000 to Mar: 2015. AFT model results are reported for the full sample across thirty-two countries and four sub-regions used in this study. We control for the firm–level industry and year fixed effects using dummy variables. 'Observations' is the number of observations used in each regression model. For the detailed definition and construction of the proxy variables, please refer to the Appendix A1.

	Panel A: Firm-level characteristics and level of market development									
	Full Sample		North-America		Europe		BRICS		East-Asia & Aust.	
Variables	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.
Day one ret.	0.0007	(0.1416)	0.0016	(0.4050)	0.0020	(0.0319)	0.0018	(0.2495)	0.0002	(0.8497)
Ln_Mkt Cap.	0.0190	(0.0940)	0.0993	(0.0024)	0.1164	(0.0001)	0.0884	(0.0901)	0.0743	(0.0057)
EBIT to TA	0.0041	(0.0263)	0.0101	(0.0230)	0.0057	(0.0022)	0.0082	(0.0675)	0.0001	(0.0597)
Sales to TA	0.0009	(0.0246)	0.0017	(0.0722)	0.0001	(0.8528)	0.0017	(0.0703)	0.0010	(0.0957)
MTB	-0.0096	(0.0449)	0.0208	(0.3105)	0.0023	(0.6795)	-0.0288	(0.3097)	0.0036	(0.6537)
Debt to TA	-0.0007	(0.0075)	-0.0018	(0.3766)	-0.0003	(0.1086)	-0.0051	(0.5119)	-0.0032	(0.0018)
Growth_TA	-0.0005	(0.1138)	0.0002	(0.8321)	-0.0003	(0.5056)	0.0015	(0.2973)	-0.0007	(0.2754)
GDPpcG	0.0995	(0.0000)	0.0296	(0.5954)	0.0192	(0.6753)	0.0254	(0.7673)	0.0777	(0.0000)
Turnover	0.0041	(0.0000)	0.0097	(0.0865)	0.0052	(0.0000)	0.0057	(0.0946)	0.0038	(0.0000)
StTraVal	0.0019	(0.0013)	0.0123	(0.0023)	0.0041	(0.0000)	0.0055	(0.1611)	0.0007	(0.1092)
Constant	2.9463	(0.0000)	3.0169	(0.0000)	2.4522	(0.0000)	5.2109	(0.0000)	3.4405	(0.0000)
Year FE	Yes		Yes		Yes		Yes		Yes	
Industry FE	Yes		Yes		Yes		Yes		Yes	
Likelihood	-5417		-860		-1690		-401		-2299	
Observations	7627		1106		1694		1245		3582	

	Panel B: Firm-level characteristics and governance factors									
	Full Sample		North-America		Europe		BRICS		East-Asia & Aust.	
Variables	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.
Day one ret.	0.0011	(0.0744)	0.0021	(0.2279)	0.0015	(0.0582)	0.0024	(0.1677)	0.0001	(0.9113)
Ln_Mkt Cap.	0.0067	(0.0689)	0.1419	(0.0000)	0.1023	(0.0000)	0.0309	(0.0630)	0.0664	(0.0006)
EBIT to TA	0.0057	(0.0000)	0.0094	(0.0144)	0.0028	(0.0932)	0.0074	(0.0730)	0.0020	(0.0628)
Sales to TA	0.0007	(0.0917)	0.0009	(0.4040)	0.0001	(0.8788)	0.0010	(0.4088)	0.0006	(0.3833)
MTB	-0.0133	(0.0019)	0.0208	(0.3126)	0.0023	(0.6509)	-0.0421	(0.1612)	-0.0054	(0.4872)
Debt to TA	-0.0006	(0.0631)	-0.0009	(0.6165)	-0.0002	(0.5837)	-0.0012	(0.8974)	-0.0038	(0.0009)
Growth_TA	-0.0005	(0.1376)	0.0003	(0.7430)	-0.0003	(0.6310)	0.0014	(0.3769)	-0.0006	(0.3866)
Bondholder right	-0.1334	(0.0000)	-0.0002	(0.2214)	-0.1188	(0.0001)	-1.1474	(0.0128)	-0.1371	(0.1653)
Shareholder right	0.1756	(0.0021)	1.3124	(0.0000)	0.2019	(0.0000)	0.1464	(0.0892)	0.1913	(0.0041)
Constant	4.4938	(0.0000)	8.6305	(0.0000)	3.5990	(0.0000)	2.8589	(0.0000)	5.2288	(0.0000)
Year FE	Yes		Yes		Yes		Yes		Yes	
Industry FE	Yes		Yes		Yes		Yes		Yes	
Likelihood	-5440		-860		-1660		-398		-2256	
Observations	7627		1106		1694		1245		3582	

Table 6: IPO characteristics, level of market development and stakeholder protection in determining the IPO survival.

This table reports the baseline regression results for the IPO listing-time firm-level characteristics, level of market development, and stakeholder protection level of the market / stock exchange where the IPO firm decides to list. We use the AFT model to study the determinants of the survival rate of the newly listed IPOs between Jan: 2000 and Dec: 2008, tracked from Jan: 2000 to Mar: 2015. AFT model results are reported for the full sample across thirty-two countries and four sub-regions used in this study. We control for the firm–level industry and year fixed effects using dummy variables. 'Observations' is the number of observations used in each regression model. For the detailed definition and construction of the proxy variables, please refer to the Appendix A1.

	Full Sample		North-America		Europe		BRICS		East-Asia & Aust.	
Variables	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.
Day one ret.	0.0007	(0.1757)	0.0017	(0.3476)	0.0015	(0.0490)	0.0028	(0.0604)	0.0001	(0.8784)
Ln_Mkt Cap.	0.0123	(0.3988)	0.1395	(0.0000)	0.0855	(0.0001)	0.0071	(0.9264)	0.0922	(0.0007)
EBIT to TA	0.0040	(0.0119)	0.0091	(0.0308)	0.0028	(0.0882)	0.0065	(0.0266)	0.0011	(0.0703)
Sales to TA	0.0009	(0.0289)	0.0012	(0.2414)	0.0001	(0.9724)	0.0017	(0.1571)	0.0006	(0.3870)
MTB	-0.0092	(0.0296)	-0.0203	(0.3148)	0.0036	(0.4746)	-0.0430	(0.1272)	0.0041	(0.6300)
Debt to TA	-0.0006	(0.0460)	-0.0013	(0.4883)	-0.0002	(0.4257)	-0.0025	(0.7704)	-0.0035	(0.0012)
Growth_TA	-0.0005	(0.1050)	-0.0002	(0.7940)	-0.0002	(0.6439)	-0.0014	(0.3598)	-0.0007	(0.3338)
GDPpcG	0.1060	(0.0000)	0.0330	(0.7569)	-0.0029	(0.9481)	0.0930	(0.4459)	0.0782	(0.0000)
Turnover	0.0030	(0.0000)	0.0099	(0.0323)	0.0035	(0.0000)	0.0070	(0.0921)	0.0026	(0.0557)
StTraVal	0.0009	(0.0438)	0.0094	(0.0079)	0.0005	(0.5153)	0.0078	(0.0955)	0.0023	(0.0063)
Bondholder right	-0.2121	(0.0000)	-0.0001	(0.2241)	-0.1298	(0.0000)	-0.9926	(0.0227)	-0.3765	(0.0006)
Shareholder right	0.0230	(0.0613)	0.8419	(0.0068)	0.2188	(0.0000)	0.2526	(0.0456)	0.0734	(0.0566)
Constant	3.6383	(0.0000)	6.7617	(0.0000)	3.4322	(0.0000)	1.6473	(0.1484)	4.1991	(0.0000)
Year FE	Yes		Yes		Yes		Yes		Yes	
Industry FE	Yes		Yes		Yes		Yes		Yes	
Likelihood	-5372		-855		-1654		-395		-2225	
Observations	7627		1106		1694		1245		3582	

Table 7: Robustness check on the significance of agency and disclosure issues in determining the IPO survival. This table reports the baseline regression results for the IPO listing-time firm-level characteristics, level of market development, and stakeholder protection level of the market / stock exchange where the IPO firm decides to list. Besides the regular IPO listing-time characteristics, we also incorporate extra proxies for listing-time agency issues and regulatory disclosure requirements in the regression model. We use the AFT model to study the determinants of the survival rate of the newly listed IPOs between Jan: 2000 and Dec: 2008, tracked from Jan: 2000 to Mar: 2015. AFT model results are reported for the full sample across thirty-two countries. We control for the firm–level industry and year fixed effects using dummy variables. 'Observations' is the number of observations used in each regression model. For the detailed definition and construction of the proxy variables, please refer to the Appendix A1.

	Model I: Full Sample		Model II: Full Sample		Model III: Full Sample		Model IV: Full Sample	
Variables	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.
Day one ret.	0.0006	(0.2496)	0.0002	(0.7353)	0.0005	(0.2841)	0.0003	(0.6091)
Ln_Mkt Cap.	0.0326	(0.0494)	0.0180	(0.3953)	0.0070	(0.6739)	0.0525	(0.0049)
EBIT to TA	0.0031	(0.0171)	0.0039	(0.0504)	0.0047	(0.0096)	0.0027	(0.0189)
Sales to TA	0.0001	(0.6451)	0.0007	(0.1652)	0.0009	(0.1528)	0.0006	(0.0792)
MTB	-0.0019	(0.7082)	-0.0044	(0.3925)	-0.0116	(0.0045)	-0.0005	(0.9155)
Debt to TA	-0.0004	(0.0678)	-0.0005	(0.0189)	-0.0003	(0.2247)	-0.0005	(0.0184)
Growth_TA	-0.0004	(0.2346)	-0.0006	(0.1423)	-0.0006	(0.1640)	-0.0005	(0.1490)
GDPpcG	0.1015	(0.0000)	0.1182	(0.0000)	0.1153	(0.0000)	0.0460	(0.0930)
Turnover	0.0014	(0.0710)	0.0026	(0.0001)	0.0032	(0.0004)	0.0009	(0.0211)
StTraVal	0.0001	(0.9960)	0.0009	(0.1718)	0.0013	(0.0518)	0.0021	(0.0001)
Bondholder right	-0.2243	(0.0000)	-0.3334	(0.0000)	-0.2059	(0.0000)	-0.1307	(0.0008)
Shareholder right	0.0939	(0.0501)	0.0787	(0.0065)	0.0116	(0.0026)	0.1153	(0.0616)
Ln_Age	0.2134	(0.0000)						
Insider own			0.0034	(0.0418)				
Cash to TA					0.0035	(0.0706)		
Acc. Disc.							-0.0649	(0.0000)
Constant	3.8806	(0.0000)	3.2102	(0.0000)	3.3694	(0.0000)	7.9297	(0.0000)
Year FE	Yes		Yes		Yes		Yes	
Industry FE	Yes		Yes		Yes		Yes	
Likelihood	-4237		-3339		-4221		-5002	
Observations	6391		4491		6209		6813	

Appendix A1: List of variables used in this study and their respective definitions and source

Variables / acronym	Definition
Initial return – Day one ret.	Logarithmic return of first – day secondary market closing price divided by the offer price, in percentage. *Source: SDC Platinum Database, Datastream / Worldscope.*
Market capitalization – Mkt. cap	Listing-time market capitalization *i.e.* the offer size of the IPO firm, in million US$. In AFT model we use the natural logarithm of the market cap. *Source: Datastream / Worldscope.*
Total assets – TA	Listing-time total assets of the firm, in million US$. *Source: Datastream / Worldscope.*
Earnings ratio – EBIT to TA	Earnings before interest but after tax divided by total assets for the financial year when the IPO firm is listed, in percentage. *Source: Datastream / Worldscope.*
Sales ratio – Sales to TA	Total sales divided by total assets for the financial year when the IPO firm is listed, in percentage. *Source: Datastream / Worldscope.*
Market to book – MTB	Market value of equity divided by the book value of equity at the time of the IPO firm listing. *Source: SDC Platinum Database, Datastream / Worldscope.*
Leverage ratio – Debt to TA	The sum of short-term and long-term debt divided by the total assets for the financial year when the IPO firm is listed, in percentage. *Source: Datastream / Worldscope.*
Total assets growth – Growth_TA	The relative change / growth in total assets for the financial year when the IPO firm is listed compared to financial year prior listing, in percentage. *Source: Datastream / Worldscope.*
Firm age - Age	Age of the firm, in years, at the time of listing since it was founded / incorporated. In AFT model we use the natural logarithm of age. *Source: SDC Platinum Database / Worldscope / Amadeus Osiris.*
Ownership concentration – Insider	The number of shares held by the insiders (shareholders who hold 5% or more of the outstanding shares, such as managers, officers, directors, immediate families, other firms or individuals) as a percentage of the total number of outstanding common shares, at the time of listing of the IPO firm. *Source: SDC Platinum Database / Worldscope.*
Cash holding – Cash to TA	The sum of cash and short-term investments divided by the total assets for the financial year when the IPO firm is listed, in percentage. *Source: Datastream / Worldscope.*
GDP Growth – GDPpcG	GDP per capita growth of the country for the financial year in which the IPO firm is listed, in percentage. *Source: World Banks World Development Indicators.*
Turnover ratio – Turnover	Total value of shares traded in a financial year divided by the average domestic market cap for the same financial year, normalized by the GDP of the country for the year in which the IPO firm is listed, in percentage. *Source: World Banks World Development Indicators.*
Stocks traded value – StTraVal	Total value of shares traded in a financial year, normalized by the GDP of the country for the year in which the IPO firm is listed, in percentage. *Source: World Banks World Development Indicators.*
Creditor rights – Bondholder right	An index aggregating creditor rights. A score of one is assigned when each of the following rights of secured lenders is defined in laws and regulations: First, there are restrictions, such as creditor consent or minimum dividends, for a debtor to file for reorganization. Second, secured creditors are able to seize their collateral after the reorganization petition is approved, *i.e.* there is no *automatic stay* or *asset freeze*. Third, secured creditors are paid first out of the proceeds of liquidating a bankrupt firm, as opposed to other creditors such as government or workers. Finally, if management does not retain administration of its property pending the resolution of the reorganization. The index ranges from 0 (weak creditor rights) to 4 (strong creditor rights*). Source: Djankov et al., (2007).*
Anti-director minority shareholder rights – Shareholder right	Aggregate index of shareholder rights. The index is formed by summing: (1) vote by mail; (2) shares not deposited; (3) cumulative voting; (4) oppressed minority; (5) pre-emptive rights; and (6) capital to call a meeting. The index ranges from 0 (weak shareholder rights) to 6 (strong shareholder rights*). Source: Djankov et al. (2008).*
Accounting disclosure – Acc. disc.	Accounting disclosure measures the proportion of 85 financial disclosures included in a representative sample of companies annual reports, and is commonly used by researchers to capture a variety of country-level financial reporting characteristics. The set of disclosures include specific items from the income statement, balance sheet, and funds flow statement, as well as more general items related to accounting standards. *Source: Bhattacharya et al. (2003).*
North-America	Canada, The U.S.A.
Europe	Austria, Belgium, Denmark, Finland, France, Germany, Greece, Italy, The Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, The U.K.,
BRICS	Brazil, China, India, Russian Federation, South Africa
East-Asia & Aust.	Australia, Hong Kong, Japan, Malaysia, New Zealand, Singapore, South Korea, Taiwan, Thailand

Appendix A2: Determinants of IPO survival using Cox Proportional Hazard model.

This appendix reports the Cox Proportional Hazard model results for the determinants of the survival rate of the newly listed IPOs between Jan: 2000 and Dec: 2008, tracked from Jan: 2000 to Mar: 2015. Determinants used in this test are the IPO listing-time firm-level characteristics, level of market development, and stakeholder protection level of the market / stock exchange where the IPO firm decides to list. Cox Proportional Hazard model results are reported for the full sample across thirty-two countries and four sub-regions used in this study. We control for the firm–level industry and year fixed effects using dummy variables. 'Observations' is the number of observations used in each regression model. For the detailed definition and construction of the proxy variables, please refer to the Appendix A1.

	Full Sample		North-America		Europe		BRICS		East-Asia & Aust.	
Variables	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.
Day one ret.	-0.0006	(0.2790)	-0.0018	(0.3261)	-0.0012	(0.0815)	-0.0024	(0.1977)	-0.0001	(0.9108)
Ln_Mkt Cap.	-0.0052	(0.7416)	-0.1132	(0.0005)	-0.1158	(0.0000)	-0.0095	(0.8958)	-0.0830	(0.0003)
EBIT to TA	-0.0041	(0.0098)	-0.0081	(0.0439)	-0.0019	(0.0721)	-0.0053	(0.0634)	-0.0008	(0.0810)
Sales to TA	-0.0010	(0.0076)	-0.0011	(0.2622)	-0.0001	(0.9865)	-0.0015	(0.2717)	-0.0009	(0.1874)
MTB	0.0076	(0.0826)	0.0207	(0.4114)	0.0036	(0.5483)	0.0451	(0.0480)	-0.0050	(0.5410)
Debt to TA	0.0004	(0.0181)	0.0017	(0.4368)	0.0002	(0.3473)	0.0036	(0.7134)	0.0027	(0.0103)
Growth_TA	0.0005	(0.1101)	0.0003	(0.7564)	0.0002	(0.9392)	0.0021	(0.2289)	0.0008	(0.1552)
GDPpcG	-0.1086	(0.0000)	0.0381	(0.5988)	-0.0089	(0.8341)	-0.0985	(0.4371)	-0.0829	(0.0000)
Turnover	-0.0026	(0.0068)	0.0093	(0.0352)	-0.0041	(0.0001)	-0.0062	(0.0776)	-0.0028	(0.0938)
StTraVal	-0.0003	(0.4511)	0.0094	(0.0032)	-0.0006	(0.5447)	-0.0072	(0.1453)	-0.0029	(0.0079)
Bondholder right	0.2210	(0.0000)	0.0001	(0.7845)	0.1593	(0.0000)	0.9670	(0.0284)	0.3987	(0.0000)
Shareholder right	-0.0354	(0.0455)	-0.7062	(0.0185)	-0.1886	(0.0000)	-0.2807	(0.0180)	-0.1241	(0.0945)
Year FE	Yes		Yes		Yes		Yes		Yes	
Industry FE	Yes		Yes		Yes		Yes		Yes	
Likelihood	-14592		-1866		-4601		-631		-5270	
Observations	7627		1106		1694		1245		3582	

Appendix A3: Determinants of domestic only IPO listings.

This appendix reports the baseline regression results for the domestic only IPO firm's listing-time firm-level characteristics, level of market development, and stakeholder protection level of the market / stock exchange where the IPO firm decides to list. We use the AFT model to study the determinants of the survival rate of the newly listed IPOs between Jan: 2000 and Dec: 2008, tracked from Jan: 2000 to Mar: 2015. Here we drop all the IPOs which were cross listed in more than one stock market at the time of listing or any time in future during our sample period, IPOs which were delisted due to M&A activity or the IPO firms which decided to have an ADR besides the domestic stock market listing at the time of IPO or any time in future during our sample period. AFT model results are reported for the full sample across thirty-two countries and four sub-regions used in this study. We control for the firm–level industry and year fixed effects using dummy variables. 'Observations' is the number of observations used in each regression model. For the detailed definition and construction of the proxy variables, please refer to the Appendix A1.

	Full Sample		North-America		Europe		BRICS		East-Asia & Aust.	
Variables	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.	Co-eff.	P-Val.
Day one ret.	0.0013	(0.1143)	0.0010	(0.6409)	0.0014	(0.1996)	0.0064	(0.0001)	0.0002	(0.9155)
Ln_Mkt Cap.	0.0621	(0.0001)	0.1081	(0.0004)	0.1728	(0.0001)	0.0456	(0.4547)	0.0312	(0.5205)
EBIT to TA	0.0044	(0.0409)	0.0089	(0.0944)	0.0028	(0.0843)	0.0146	(0.0878)	0.0019	(0.0710)
Sales to TA	0.0021	(0.0001)	0.0037	(0.0000)	0.0003	(0.6778)	0.0025	(0.2452)	0.0018	(0.0111)
MTB	-0.0098	(0.3792)	-0.0129	(0.5731)	-0.0069	(0.5403)	-0.0451	(0.0845)	-0.0001	(0.9936)
Debt to TA	-0.0007	(0.0136)	-0.0036	(0.2280)	-0.0005	(0.0218)	-0.0022	(0.6876)	-0.0047	(0.0354)
Growth_TA	-0.0002	(0.5467)	-0.0012	(0.3816)	-0.0002	(0.7476)	-0.0020	(0.2707)	-0.0008	(0.5668)
GDPpcG	0.0678	(0.0011)	0.0346	(0.6255)	-0.0394	(0.4405)	0.1664	(0.2352)	0.0283	(0.0740)
Turnover	0.0038	(0.0000)	0.0024	(0.0659)	0.0034	(0.0008)	0.0061	(0.0714)	0.0008	(0.0616)
StTraVal	0.0024	(0.0000)	0.0045	(0.3108)	0.0011	(0.1421)	0.0121	(0.0556)	0.0048	(0.0000)
Bondholder right	-0.2312	(0.0000)	0.0001	(0.5547)	-0.1154	(0.0241)	-0.3564	(0.4247)	-0.4947	(0.0043)
Shareholder right	0.0222	(0.0763)	0.9013	(0.0030)	0.2143	(0.0007)	0.3632	(0.0566)	0.2904	(0.0570)
Constant	3.8585	(0.0000)	6.8878	(0.0000)	3.4490	(0.0000)	2.9254	(0.0723)	6.3106	(0.0000)
Year FE	Yes		Yes		Yes		Yes		Yes	
Industry FE	Yes		Yes		Yes		Yes		Yes	
Likelihood	-3177		-523		-1028		-259		-1163	
Observations	5739		848		1157		983		2751	